

Forecasting Daily Gold Futures Prices with Classical Time-Series Methods and Macro-Financial Predictors

Team members

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1 Motivation and research question

Gold is a globally traded asset often viewed as a hedge against inflation, uncertainty and currency movements. Because gold futures are liquid and trade daily on regulated exchanges, they provide a realistic time-series forecasting problem with clear evaluation criteria. This project aims to determine **how well we can forecast one-step-ahead daily gold futures prices using classical methods from Stat 443**, and to assess **whether incorporating macro-financial predictors improves forecast accuracy relative to purely univariate models**.

The primary response variable will be the **daily closing price of gold futures**. We will also examine log-returns for robustness. Beyond this main question we may explore whether external predictors have larger impacts during high-volatility periods and whether technical indicators derived from gold prices provide additional predictive power once macro variables are included.

2 Data description and sources

To satisfy the requirement to compare models **with and without external predictors**, we will assemble a dataset that combines daily gold futures prices with macro-financial variables. The core gold series comes from an existing Kaggle dataset, and the exogenous variables are obtained from publicly available sources such as Yahoo Finance and the Federal Reserve's FRED database. All series will be aligned to a common daily trading calendar, with missing values handled carefully (e.g., only using lagged values available at forecast time).

Gold futures series (response variable)

- We will use the Kaggle "Gold Price Forecasting Dataset" (ticker GC=F). The dataset provides daily open, high, low, close and volume for gold futures over roughly the last five years and includes several technical indicators such as moving averages, rolling volatilities, and RSI.
- The series is sufficiently long to fit ARIMA and ETS models and to perform a training-holdout split. However, because the dataset spans only ~five years, it may not capture longer-term cycles or structural breaks. We will mitigate this by focusing on short-term (one-step-ahead) forecasts.

External predictors

To allow for ARMAX/regression models, we plan to augment the gold series with a few daily macro-financial variables that may influence gold prices:

- **U.S. Dollar Index:** an index measuring the value of the U.S. dollar against a basket of major currencies; a stronger dollar typically implies lower gold prices.
- **S&P 500 index:** equity market sentiment in the US can affect demand for gold.
- **WTI crude oil price:** a benchmark for oil prices; energy prices shape inflation expectations and thus demand for gold.

- **10-year U.S. Treasury yield:** a benchmark interest rate that affects the opportunity cost of holding non-yielding assets like gold.

These predictors will be lagged appropriately to ensure no look-ahead bias when forecasting.

3 Planned modelling and forecasting approach

3.1 Forecasting setup

1. **Dataset preparation:** Merge the gold futures series with the external predictors on trading days. Handle weekends and holidays by retaining only days when gold trades; for each gold observation, ensure predictor values are lagged appropriately. Convert prices to log-returns where necessary and explore the need for differencing to achieve stationarity.
2. **Train-test split:** Divide the data chronologically into a training set (e.g., the first ~80 % of observations) and a holdout set (the remaining ~20 %). Use the training set for model fitting and selection, and use the holdout set for rolling one-step-ahead forecast evaluation.
3. **Evaluation metrics:** For each method, compute the root-mean-square (RMS) error of the one-step-ahead forecasts on the holdout set. Also report in-sample RMS prediction error (using one-step-ahead predictions on the training set) for context. Rank models by out-of-sample RMS error. Include baseline methods as required - naïve persistence and training-set average forecasts.

3.2 Model classes

1. **Exponential smoothing (ETS):** Forecast the series using simple and damped exponential-smoothing models.
2. **ARIMA:** Fit appropriate ARMA/ARIMA models to the gold series and generate one-step-ahead forecasts.
3. **Regression models with external variables:** Build regressions of the gold price on lagged macro predictors; consider versions with ARMA errors (ARMAX).

3.3 Diagnostics and robustness

For each model, perform residual diagnostics—check for autocorrelation and heteroscedasticity—and adjust models where needed. Compare performance on both the level series and log-returns to ensure robustness against non-stationarity.

4 Evaluation and comparison

We will compare models using the out-of-sample root-mean-square error of one-step-ahead forecasts and also report in-sample RMS errors for context. Residual diagnostics will be summarized to ensure models are not overfitting and that assumptions are reasonably satisfied. We will present results in tables and plots and discuss which models perform best.

5 Data source links

- **Gold futures series (GC=F):** <https://www.kaggle.com/datasets/vishardmehta/gold-price-forecasting-dataset>
- **U.S. Dollar Index:** <https://fred.stlouisfed.org/series/DTWEXBGS>
- **S&P 500 index:** <https://fred.stlouisfed.org/series/SP500>
- **WTI crude oil:** <https://fred.stlouisfed.org/series/DCOILWTICO>
- **10-year U.S. Treasury yield:** <https://fred.stlouisfed.org/series/DGS10>